

Muhammad Rohaan Qutab

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PROFILE

Computer Science undergraduate with strong interest in Python development, cloud computing, and machine learning. Actively building projects in Python, exploring ML concepts, and working on practical applications. Seeking internship opportunities to apply and improve technical and problem-solving skills.

SKILLS

Programming: Python, C++ (basic-intermediate)

Core CS: Data Structures, OOP, Problem Solving, Basic AI Algorithms

Data Analysis / ML (Learning): NumPy, Pandas, Basic Machine Learning

Tools: Git, GitHub, VS Code, Jupyter Notebook

Other: Basic Linux, Command Line, Debugging

PROJECTS

n8n Automation Projects

- Built workflow automation systems using n8n for inventory analysis, LinkedIn post generation, and email prioritization.
- Automated LinkedIn content creation and publishing based on audience and genre inputs.
- Developed an email classification system to separate urgent and non-urgent emails for faster handling.

NeuroMap – Brain Tumor Detection and Classification (Web App)

- Built a web app that accepts brain MRI uploads and classifies tumors into four categories using a custom CNN trained from scratch (no pretrained models).
- Displays prediction confidence scores and a Grad-CAM heatmap overlay highlighting the region the model focused on.
- **Tech Stack:** Python, TensorFlow/Keras, Grad-CAM, Streamlit

Python Data Analysis Mini Projects (NumPy / Pandas / Matplotlib)

- Performed exploratory data analysis on structured datasets.
- Cleaned and transformed data using Pandas.
- Generated insights using aggregation and filtering techniques.

Bird Detection and Intrusion Monitoring System (Computer Vision)

- Built a real-time bird detection and tracking system using YOLOv8, OpenCV, and Python for live video streams.
- Implemented object tracking with unique IDs and restricted-zone intrusion detection.
- Developed real-time event-based alerts for bird entry into predefined restricted areas.

Insurance Cost Prediction (ML Project)

- Built a Linear Regression model to predict insurance charges using features like age, BMI, number of children, and smoking status.
- Achieved R^2 score ≈ 0.78 and MAE ≈ 4000 on test data.
- Developed a Streamlit app for real-time predictions. **Tech Stack:** Python, Pandas, Scikit-learn, Streamlit.

Hypertension Risk Prediction System (C++ | OOP + ML Simulation)

- Built a C++ system to predict hypertension risk using Logistic Regression on patient data.
- Implemented feature scaling, sigmoid function, and gradient descent-based training.
- Classified patients into Low, Medium, and High risk groups based on predicted probability.

EDUCATION

University of Central Punjab

Bachelor of Science in Computer Science

Lahore, Pakistan

2023 – Expected 2027

- Relevant coursework: Programming Fundamentals, Object Oriented Programming, Data Structures, Database Systems, Programming for AI, Artificial Intelligence, Cyber Security.

CERTIFICATIONS

- Python for Everybody — Coursera
- Introduction to Frontend — Meta (Coursera)
- Introduction to Cybersecurity — Cisco Netacad